

Chemistry and Chemical Biology
CHEM 2Y03
Elemental Chemistry
Winter 2027



SCIENCE

Instructor Information



Dr. Jim Ghoshdastidar
Email: ghoshda@mcmaster.ca
Office: ABB 142A

Course Information

Course Description

An engaging exploration of the role chemical elements play in our daily lives, our past, present and future. This course surveys the basic building blocks of matter and their applications in health, industry, food, technology and the environment. Three lectures, one term.

Prerequisite(s): None. CHEM 2Y03 is suitable for students of all educational backgrounds.

Anti-requisite(s): None.

Future Courses: None.

Registration Note: Students wishing to take further courses in Chemistry are invited to enrol in CHEM 1A03/1AA3.

Distance Offering: This course qualifies for students to write the tests and final exam off site if you live **more than 160km** away from the McMaster Campus during the course duration. Students who qualify and want to write the final at an offsite location in-person need to apply for the **distance section**. Application and details can be found at [SciOnline - Student Resources](#)

Student must submit the application **before the last day to add a class**, no changing sections will be approved after the last day to add a class. Any

questions about this option can be directed to Jacqueline Hook at scionline@mcmaster.ca.

Course Format

Delivery	Format	Location	Office and Advice Hours
Online Lectures	Asynchronous Online Modules	Avenue to Learn	TBA

Online Lectures will be delivered through pre-recorded asynchronous online modules to provide students with maximum flexibility. You can engage with the modules at a time and pace that suits you best. Gentle guidelines and online quizzes will help you keep track of course content and prepare for higher-weight assessments.

Virtual Office and Advising Hours are available using Microsoft Teams and are booked in 10-minute time slots using Microsoft Bookings. Choose any available Advising Hour timeslot and indicate that your session is for CHEM 2Y03.

Assessments, three tests and one final exam, will be held with in-person invigilation on designated test days and times. You can choose to complete your assessments: 1) On-Campus at McMaster with in-person invigilation (at no cost) or Off-Campus with an in-person invigilator/proctor (an additional fee will apply). See the registration note on Distance Invigilation for more information.

Important Links

- [Mosaic](#)
- [Avenue to Learn](#)
- [Student Accessibility Services - Accommodations](#)
- [McMaster University Library](#)
- [eReserves](#)
- [Open Education Resources](#)

Course Learning Outcomes

- Analyze how the chemical elements have influenced human history, culture, industry, and scientific innovation.
- Recognize the interdisciplinary nature of chemistry and its contributions to the fields of medicine, engineering, the arts, and modern technology.

- Critically evaluate scientific and popular media claims related to the chemical elements and their uses
- Interpret and communicate information about the chemical elements using appropriate scientific terminology, data, and evidence.

Required Materials and Texts

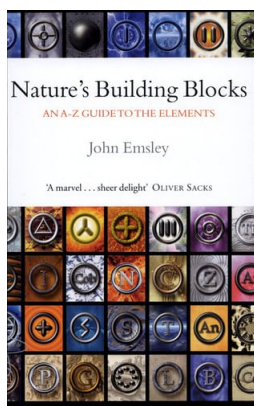
Please sign in with your MacID [here](#) to view your booklist

There are no required texts for this course.

A list of course materials for your interest and further reading are provided under optional course materials.

Optional Course Materials

Please sign in with your MacID [here](#) to view your booklist



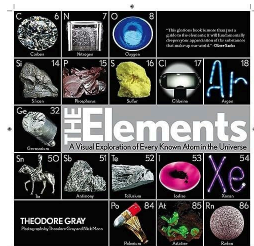
Nature's Building Blocks An A-Z Guide to the Elements

ISBN: 9780198503408

Authors: John Emsley

Publisher: Oxford University Press

Publication Date: 2003



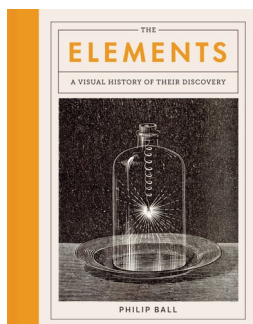
The Elements A Visual Exploration of Every Known Atom in the Universe

ISBN: 9781579128142

Authors: Theodore W. Gray

Publisher: Black Dog & Leventhal Publishers

Publication Date: 2009



The Elements A Visual History of Their Discovery

ISBN: 9780226775951

Authors: Philip Ball

Publisher: University of Chicago Press

Publication Date: 2021

Class Format

Online

Lecture Format and Resources

The online asynchronous lectures and course notes are the best resources to inform yourself about the topics emphasized in the course.

Course notes and online asynchronous lectures will be available as pdf files on Avenue to Learn before the scheduled week of the lecture. The notes follow the lectures and contain most of the text shown in the pre-lectures or in-person lectures. However, portions of the text have been omitted; I encourage you to add notes as you engage with the lectures.

The asynchronous online lectures will consist of short video lectures that cover a specific element of the course material. Please refer to these modules as required. Students are encouraged to regularly and fully engage with all course materials to achieve mastery.

In addition to the asynchronous lectures, you can access the Avenue to Learn Discussion Board and one-on-one Microsoft Teams office hour appointments to clarify concepts.

Course Evaluation

Final grades for CHEM 2Y03 will automatically be calculated using the following grading scheme.

Evaluation Component	Grading Scheme
Asynchronous A2L Lectures (Drop 6)	15%
3 Minute Lecture Presentation	15%

6 A2L Online Quizzes (Drop 1)	15%
Tests	30%
Final Exam	25%
Total:	100%

The following criteria must be met to pass the course: (1) a final grade in the course $\geq 50\%$ or higher, and (2) the final examination must be written. There are no extra/bonus assignments following the course to boost one's grades. Your best effort throughout the course is expected. Regular rounding rules will be followed in determining your final grade – a grade of 49.50 will be rounded to 50, and 49.49 will be truncated to 49.

Course Evaluation Details

Final Exam

A 2-hour cumulative multiple-choice In-Person Final Examination will be scheduled during the Final Exam Period. The Final Examination will consist of two parts: Part 1 is essentially Test 4 which will consist of material from Units 10 to 12 (10%) and Part 2 which is cumulative and draws material from the entire course (15%).

You can choose to complete your Final Exam: 1) On-Campus at McMaster with in-person invigilation (at no cost) or Off-Campus with an in-person invigilator/proctor (an additional fee will apply) if you live more than 160 km from campus. See the registration note on Distance Invigilation for more information.

All McMaster rules and procedures relating to Academic Dishonesty and Academic Integrity apply to all tests and examinations; all violations will result in a penalty. You must do your own work. Any communication with classmates or third parties regarding any aspect of the Examination is prohibited once you begin the exam.

Tests

This course will hold three 50-minute, in-person multiple-choice tests. Should you miss a test, the weight of the test will be shifted to the cumulative portion (Part 2) of the Final Exam. No MSAFs are required for missing a test.

You can choose to complete your Final Exam: 1) On-Campus at McMaster with in-person invigilation (at no cost) or Off-Campus with an in-person invigilator/proctor (an additional fee will apply) if you live more than 160 km

from campus. See the registration note on Distance Invigilation for more information.

Information on the date and time that tests will be held will be available on A2L.

Online A2L Quizzes

Six quizzes are available during the term through Avenue. Each quiz will be available a minimum of seven days before its due date. All students are pre-emptively granted a one-week extension on these quizzes, except for the final quiz which will close after one week or on the last day of registrar-scheduled classes (whichever comes first). You will have two attempts at each quiz where the first attempt is the full set of questions and the second is an opportunity to rework incorrect responses for additional credit. Only the top 5 of the 6 quiz marks will be counted towards your final grade.

3 Minute Lecture Presentation

If you had one Element to add to this course, what would it be and why? Here is your chance to make a pitch to the class and me, your instructor, to add it to next year's course. Each of you will present one slide and have 3 minutes to convince your audience. This assignment will allow you to synthesize and summarize the significance of an element not covered in this course into a short and accessible talk for an undergraduate science audience. You will be evaluated on your communication, comprehension, scientific accuracy, and presentation engagement. Note that presentations are not live but recorded.

Asynchronous Lecture Modules

The asynchronous A2L online lecture modules contain lecture slides, lecture video, captions, transcriptions, and helpful links. Students will earn marks through the modules by watching/listening to the narration on each slide to completion and answering the questions embedded within the module. **The Asynchronous Lecture Modules are automatically graded by Avenue to Learn based on your highest graded attempt. No manual regrades are offered.** Please verify your asynchronous lecture module progress through Avenue to Learn grades.

Grading Policy

All grades will be recorded in the Avenue to Learn Gradebook. Any grade discrepancies must be reported within five business days of the grade being posted and work returned. It is the student's responsibility to ensure that their grades have been reported accurately.

Any requests for re-grading must be emailed within five business days of an assessment being returned to the student with a rationale for the request. If approved, the entire assessment will be regraded, which may result in an increase, decrease or no change to the original mark.

Undergraduate Grading Scale

The grades breakdown for McMaster's 12-Point Grading Scale

Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79
B	8	73-76
B-	7	70-72
C+	6	67-69
C	5	63-66
C-	4	60-62
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

Course Schedule

Week	Week Beginning on	Chapter & Lecture Topics	Online Lectur
1	Tuesday, January 5	1 – Introduction	Intro 1 – Course Syllab Intro 2 – Introduction to Intro 3 – Development Table
2	Monday, January 11	2 – The Primordial Elements	Lecture 1 – Hydrogen Lecture 2 – Helium Lecture 3 – Lithium
3	Monday, January 18	3 – Life Triad	Lecture 6 – Carbon Lecture 7 – Nitrogen Lecture 8 – Oxygen

Week	Week Beginning on	Chapter & Lecture Topics	Online Lectur
4	Monday, January 25	4 – The Earth-Life Triad	Lecture 14 – Silicon Lecture 15 – Phosphorus Lecture 16 – Sulfur
5	Monday, February 1	5 – The Alkali Metals	Test 1 – Units 1 to 3 Lecture 11 – Sodium Lecture 19 – Potassium
6	Monday, February 8	6 – The Halogens	Lecture 9 – Fluorine Lecture 17 – Chlorine Lecture 53 – Iodine
-	Monday, February 16	Reading Week	
7	Monday, February 22	7 – The Alkaline Earth Metals	Test 2 – Units 4 to 6 Lecture 12 – Magnesium Lecture 20 – Calcium
8	Monday, March 1	8 – Ancient Metals	Lecture 26 – Iron Lecture 29 – Copper Lecture 50 – Tin
9	Monday, March 8	9 – Precious Metals	Lecture 47 – Silver Lecture 78 – Platinum Lecture 79 – Gold
10	Monday, March 15	10 – Structural Metals	Lecture 13 – Aluminum Lecture 22 – Titanium Lecture 28 – Nickel
11	Monday, March 22	11 – Toxic Elements	Test 3 – Units 7 to 9 Lecture 33 – Arsenic Lecture 80 – Mercury Lecture 82 – Lead
12	Monday, March 29	12 – Nuclear Three	Lecture 88 – Radium Lecture 92 – Uranium Lecture 94 – Plutonium
13	Monday, Apr 5	13 – The Future	March Madness – 3ML
13	Thursday, Apr 9 to	CHEM 2Y03 Test 4 and Final Exam	
14	Wednesday April 22		

* Please note that this schedule and the number of modules is approximate and subject to change.

Absences, Missed Work, Illness

Academic Absences and Consideration

Requests for Relief for Missed Academic Term Work

If you are absent from the University for a minor medical reason lasting fewer than 3 days, you may report your absence without documentation using the McMaster Student Absence Form (MSAF). Absences for a longer duration or for other reasons must be reported to your Faculty/Program office with documentation, and relief from term work may not necessarily be granted.

[McMaster Student Absence Form \(MSAF\)](#): In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar, "Requests for Relief for Missed Academic Term Work."

When using the MSAF, report your absence to me using the following email address: ghoshda@mcmaster.ca. Within a few days of submitting your MSAF, check your email to see that your MSAF has been acknowledged.

Academic Accommodation for Religious, Indigenous or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, Indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office **typically within 10 working days** of the beginning of the term in which they anticipate a need for accommodation or to the Registrar's Office before their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Academic Accommodation, Accessibility and Inclusion

As the instructor of this course, I endeavour to provide an inclusive learning environment. If you discover an accessibility barrier, please get in touch with me. If you experience barriers to learning in this course and may need academic accommodation to support your learning, please register with [Student Accessibility Services \(SAS\)](#).

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services \(SAS\)](#) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's [Academic Accommodation of Students with Disabilities](#) policy.

Course Material and Accessibility Features Technology	
PDF Lecture Notes	▪ Alternative text for non-decorative images ▪ Easy-to-read text contrast ▪ Reading order verification of slide text ▪ Slide titles and Hidden slide titles for all slides
Asynchronous A2L Lectures	• Single-click accessibility mode that screen readers can easily recognize. • Transcripts/notes for every slide. • Captions for audio and video narration.
Microsoft Teams	▪ Auto-generated captions for lecture recordings ▪ Fully keyboard accessibility with shortcuts for navigation ▪ Screen Reader Supported

Student Mental Health and Wellness

At McMaster University, we recognize that things can come up throughout the term that may interfere with a student's ability to succeed in their coursework. These circumstances are outside one's control and can seriously impact physical and mental well-being. Seeking help can be a challenge, especially in times of crisis.

If support is needed immediately, you can access these outside resources at any time:

- [COAST \(Crisis Outreach and Support Team\)](#) — Hamilton's 24/7 line for if you are in crisis, feeling suicidal or in need of emotional support (phone: 905-972-8338)

- [Good2Talk](#) — 24/7 hour line for post-secondary students (phone: 1-866-925-5454 or dial 211 and ask to be connected to Good2Talk)
- [Empower Me](#) — 24/7 access to confidential support through counsellors via phone at 1-833-628-5589
- [Assaulted Women's Helpline](#) – Free, 24/7, anonymous and confidential crisis counselling, safety planning and emotional support to all women who have experienced any form of abuse at 1-866-863-0511

There are additional Student Support Services that you can access on campus:

- [Counselling](#) and [Medical Care](#) — Call 905-525-9140 x 27700 to book an appointment
- [Chaplaincy Centre](#) – Call 905-525-9140 x 24207 or visit MUSC 231 for pastoral support with personal counselling and bereavement support groups.
- Natalie Lafleur, [Sexual Violence Response Coordinator](#) — 905-525-9140 ext. 20909 or email lafleurn@mcmaster.ca
- [SWHAT – Walk Safe Program](#) is a volunteer service within the McMaster Students Union that will walk or bus with students during the evening hours, 7 days a week, in all kinds of weather! Call 905-525-9140 x 27500, visit MUSC 234 or [book online](#).

I encourage all McMaster community members to access to ensure support is reachable; please start with [McMaster's Wellness Centre](#), or any of the links posted to our CHEM 2Y03 Avenue 2 Learn Course Page.

Turnitin.com

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other

software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Course Modification

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, Delta Variant... etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.

Modifications to The Course

The instructor and the University reserve the right to modify elements of the course during the semester. The University may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with students will be given with an explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and the course website weekly during the semester and to note any changes.

Generative AI: Use Prohibited

Students are not permitted to use generative AI in this course. In alignment with [McMaster academic integrity policy](#), it "shall be an offence knowingly to ... submit academic work for assessment that was purchased or acquired from another source". This includes work created by generative AI tools. Also stated in the policy is the following, "Contract Cheating is the act of "outsourcing of student work to third parties" (Lancaster & Clarke, 2016, p. 639) with or without payment." Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

APPROVED ADVISORY STATEMENTS

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](#).

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity / Plagiarism

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Courses with an On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when

they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online.**

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Academic Accommodation of Students with Disabilities

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Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

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Students requiring academic accommodation based on religious, Indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students

should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.